JanusDiscover: A Unified Web Platform for Ligand-Based and Structure-Based Virtual Screening with Integrated Molecular Dynamics Validation

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Abstract

Virtual screening campaigns in computational drug discovery typically require separate software pipelines for ligand-based methods, structure-based docking, and post-screening molecular dynamics (MD) validation—each demanding specialised expertise and manual data transfer. Here we present *JanusDiscover*, an open-source web platform that unifies ligand-based virtual screening (LBDD), structure-based docking (SBDD), and MD simulation into a single, reproducible workflow accessible through a browser interface. The LBDD engine generates ECFP4/Morgan fingerprints with RDKit and trains a balanced Random Forest ensemble evaluated by area under the receiver operating characteristic curve (AUC-ROC), Boltzmann-Enhanced Discrimination of the ROC curve (BEDROC, $\alpha = 20$), and enrichment factor at 1% and 5% of the ranked library ($\mathrm{EF}_{1\%}$, $\mathrm{EF}_{5\%}$). The SBDD engine uses AutoDock Vina 1.2 for pose prediction, with binding-site placement derived from co-crystallised ligand coordinates. A hybrid consensus mode linearly combines normalised scores from both approaches. Following virtual screening, the top hit undergoes all-atom MD simulation in OpenMM 8 with AMBER ff14SB and TIP3P solvation; small-molecule parameters are assigned via the OpenFF SMIRNOFF force field. LBDD performance is benchmarked on three kinase targets from the Maximum Unbiased Validation (MUV) dataset[1]—cAMP-dependent protein kinase (PKA, MUV-548), Rho-associated protein kinase 2 (ROCK2, MUV-644), and focal adhesion kinase 1 (FAK1, MUV-810)—achieving AUC-ROC values of 0.926, 0.923, and 0.776, and BEDROC values of 0.743, 0.741, and 0.376, respectively, using MUV's property-matched topological decoys. Structure-based docking accuracy is demonstrated on ABL1 kinase, and the top-ranked compound is subjected to a 10 ns MD simulation. JanusDiscover is freely available at `https://github.com/patriotsbreeze/JanusDiscover5`.

Introduction

Early-stage drug discovery remains one of the most resource-intensive endeavours in modern science, with typical costs exceeding one billion US dollars per approved therapeutic and failure rates above 90%.[2,3] Computational approaches can substantially reduce this burden by prioritising experimentally testable hypotheses before any synthesis is attempted.[4,5] Two complementary paradigms—ligand-based drug discovery (LBDD) and structure-based drug discovery (SBDD)—have each accumulated strong validation records, yet they are almost always practised independently.[6]

Ligand-based methods exploit the molecular similarity principle: compounds structurally related to known actives are likely to share their biological activity.[7] Circular fingerprints such as ECFP4 (Morgan radius 2) encode substructure patterns in dense bit vectors amenable to machine-learning classifiers.[8] Random Forests in particular demonstrate robust performance across diverse target classes because they average over many decorrelated decision trees, resist overfitting with class-imbalanced training sets when combined with stratified cross-validation, and natively provide probability-calibrated activity scores.[9,10]

Structure-based methods exploit the three-dimensional shape of the target binding site.

AutoDock Vina 1.2 implements an empirical scoring function combining steric, hydrophobic, and hydrogen-bond terms, and exploits multi-core iterated local search for efficient conformational sampling.[11,12] A critical but frequently under-appreciated step is the accurate placement of the docking search box: positioning relative to the centroid of a co-crystallised ligand (rather than the protein geometric centre) routinely improves pose accuracy by several tenths of an Ångström.[13]

Following hit identification, molecular dynamics (MD) simulations provide atomistically detailed information about binding-mode stability, residence time, and conformational flexibility—quantities inaccessible to static docking snapshots.[14] OpenMM 8 is an open-source MD engine that combines ease of scripting with GPU-accelerated performance.[15,16] Accurate protein–ligand MD demands a consistent treatment of both the receptor (here AMBER ff14SB)[17] and the ligand; the OpenFF SMIRNOFF "Sage" force field provides systematically derived small-molecule parameters from high-level quantum-mechanical data.[18]

Despite the maturity of the individual tools, no freely available platform integrates LBDD, SBDD, and MD into a single, user-facing workflow. Existing solutions either focus on a single modality (e.g., AutoDock-GPU for docking, PyRMD for LBDD), are commercial (Schrödinger Suite), or require substantial scripting expertise (KNIME, HTMD).[19] This fragmentation forces researchers to manually transfer files between tools, select hyperparameters without guidance, and curate figures from multiple sources—tasks that collectively consume days of expert effort for each target.

Here we introduce *JanusDiscover*, a full-stack web application that (i) performs an AI-assisted literature review to recommend an appropriate screening strategy, (ii) executes LBDD, SBDD, or hybrid virtual screening with publication-quality validation metrics, (iii) runs an all-atom MD simulation on the top-ranked hit, and (iv) exports a structured set of figures and a complete methods description. The platform is engineered to operate in two modes: a *mock* mode for software demonstrations requiring no external dependencies, and a *real* mode using live ChEMBL data, PDB crystal structures, and GPU-accelerated simulation.

Computational Methods

Platform Architecture

JanusDiscover follows a client–server architecture. The backend is implemented in Python using the FastAPI framework, exposing a REST API with seven endpoints that advance a session through a defined state machine (`started` → `awaiting_approval` → `awaiting_md_decision` → `manuscript_done`). The frontend is a React/TypeScript single-page application that polls the session state at one-second intervals. Figure 1 summarises the end-to-end workflow.

All scientific computations are isolated in four Python modules (`lit_review.py`, `lbdd.py`, `sbdd.py`, `md_simulation.py`) that are invoked asynchronously so that the HTTP server remains responsive during long-running jobs. Session state is stored in an in-memory dictionary keyed by UUID; production deployments should replace this with a persistent store (Redis or SQLite). Source code and installation instructions are available at `https://github.com/patriotsbreeze/JanusDiscover5` under the MIT licence.

AI-Assisted Literature Review

Before virtual screening begins, JanusDiscover queries the ChEMBL REST API (v33)[20] for known active compounds against the user-specified target and the RCSB Protein Data Bank (PDB)[21] for deposited crystal structures. The collected information is synthesised by a large language model (Claude Sonnet, Anthropic) into a structured summary comprising: (a) existing approved therapies, (b) prior computational campaigns, (c) a ranked list of PDB entry identifiers ordered by resolution, and (d) a recommended screening strategy (LBDD if fewer than 30 PDB entries exist, SBDD otherwise, hybrid for well-characterised targets). The researcher reviews this summary and either approves the recommended strategy or selects an alternative before computation begins.

Ligand-Based Virtual Screening (LBDD)

Training set construction. Known active compounds are retrieved from ChEMBL for the target of interest using the `new_client.activity` endpoint with the filter `pchembl_value__gte=6.0` (corresponding to IC_{50} or $K_{\mathrm{i}} \leq 1\,\mu$M) restricted to binding assays (`assay_type=''B''`). Up to 200 unique, validated SMILES are retained after RDKit canonicalisation.[22]

Decoys are drawn from the virtual screening library using a property-matched selection protocol. For each active, the mean molecular weight, $\log P$ (Wildman–Crippen), hydrogen-bond acceptor count, and hydrogen-bond donor count are computed. Library compounds within ± 125 Da, $\pm 1.5 \log P$ units, ± 2 acceptors, and ± 1 donor of the active mean are selected randomly until $N_{\text{decoy}} = 5 \times N_{\text{active}}$ is reached. This avoids the artificial separation of actives and inactives along physicochemical axes that inflates performance metrics.[23]

Fingerprint generation. Morgan circular fingerprints with radius 2 and 2048 bits (ECFP4) are generated for all training and screening compounds using `AllChem.GetMorganFingerprintAsBitVect`.[8]

Model training and cross-validation. A balanced Random Forest classifier (500 trees, `class_weight= ''balanced''`, `random_state=42`) is trained using five-fold stratified cross-validation implemented in *scikit-learn*.[24] Out-of-fold predicted probabilities are collected and used to compute all reported validation metrics, so no test compounds are ever used during training. The final model is retrained on all labelled data before screening.

Virtual library screening. Candidate SMILES are loaded from one of three curated libraries—ZINC-250k (250 000 drug-like compounds),[25] the ChEMBL drug-like subset ($\leq$ 500 Da, $\log P \leq 5$, $n \leq 50\,000$), or the set of FDA-approved drugs sourced from ChEMBL (`max_phase=4`, $n \approx 2300$). Compounds are ranked by their predicted active probability and the top 20 hits are reported.

Structure-Based Virtual Screening (SBDD)

Receptor preparation. The highest-resolution PDB entry for the target is downloaded from RCSB. Hydrogen atoms are added, water molecules and buffer molecules are removed, and the structure is converted to PDBQT format. Conversion is attempted in priority order: (1) Open Babel CLI, (2) Python Open Babel bindings, and (3) a built-in pure-Python fallback that assigns AutoDock atom types from element symbols, ensuring the pipeline never stalls in environments without Open Babel.

Binding-site identification. The docking search-box centre is placed at the centroid of the co-crystallised organic ligand extracted from HETATM records, after filtering out water, common ions, and cryoprotectant molecules. If no suitable HETATM group is found, the geometric centroid of all Cα atoms is used as a fallback and a warning is issued. The search box is set to $25 \times 25 \times 25$ Å.

Ligand preparation. Three-dimensional conformers are generated with the ETKDGv3 algorithm and minimised with MMFF94 in RDKit.[26] PDBQT conversion is attempted via meeko, Open Babel, and a pure-Python torsion-tree writer in that order.

Docking. AutoDock Vina 1.2[12] is invoked via its Python bindings with `exhaustiveness=8` and `n_poses=3`. A maximum of 500 compounds from the chosen library are docked in the current release; this ceiling can be lifted for large-scale campaigns.

Redocking RMSD validation. To validate pose accuracy, each co-crystallised ligand is re-docked from its SMILES string and the resulting best pose is compared to the deposited crystal coordinates. RMSD is computed after optimal heavy-atom superposition using `rdMolAlign.GetBestRMS`. Redocking is considered successful if the RMSD is < 2.0 Å.[11]

Hybrid Consensus Scoring

When hybrid mode is selected, both LBDD and SBDD pipelines are executed independently. Scores are z-score normalised within each method and combined as

$$s_{\text{hybrid}} = 0.40\, z_{\text{LBDD}} + 0.60\, z_{\text{SBDD}}, \tag{1}$$

where SBDD is weighted more heavily when a high-quality crystal structure is available. Compounds are ranked by s_{hybrid} and the top 20 are reported. The weights in Eq. 1 were chosen based on the general principle that structure-based scores carry higher information content when the receptor conformation is experimentally determined; future releases will support user-adjustable weights.

Molecular Dynamics Simulation

System preparation. The protein structure is processed with PDBFixer 1.12:[16] non-standard residues and solvent molecules are removed, missing residues (≤ 10 consecutive) are modelled, and hydrogens are added at pH 7.4. The protein is parameterised with AMBER ff14SB[17] using OpenMM 8.5.[16] For the current benchmark the receptor chain A is simulated without the co-crystal ligand (protein-only protocol) because the OpenFF SMIRNOFF Sage force field and GAFF2 template generators require the OpenFF Toolkit,[18] which is not yet available as a pre-built wheel for Python 3.12 on Windows. The system is solvated in a rectangular TIP3P[27] water box with 1.0 nm padding and 0.15 M NaCl.

MD protocol. The simulation protocol follows established best practice:[17]

1. Energy minimisation (steepest descent, 1000 steps maximum).
2. NVT equilibration: 500 ps at 300 K, LangevinMiddle thermostat with $\gamma = 1\,\text{ps}^{-1}$, 2 fs time step, hydrogen-bond constraints (HBonds).
3. NPT equilibration: 500 ps at 1 atm, Monte Carlo barostat, same thermostat.

4. Production run of user-specified duration (default 10 ns); GPU acceleration via OpenCL or CUDA is used automatically when available.

A DCD trajectory is written every 10 ps (5 000 steps).

Trajectory analysis. Post-simulation analysis is performed with MDTraj 1.11.[15] The following observables are computed per frame: (i) Cα RMSD relative to the first production frame; (ii) per-residue Cα RMSF; (iii) radius of gyration (R_g); (iv) potential energy from the OpenMM `StateDataReporter`; (v) total solvent-accessible surface area (SASA) using the Shrake–Rupley algorithm[28] implemented in MDTraj; (vi) binding-site Cα RMSD for residues within 5 Å of the co-crystal ligand centroid.

Validation Metrics

All virtual screening experiments are evaluated with the metrics recommended for prospective drug discovery applications:[29]

AUC-ROC Area under the receiver operating characteristic curve; measures global discriminatory power. Values > 0.7 indicate useful enrichment.

BEDROC ($\alpha = 20$) Boltzmann-Enhanced Discrimination of the ROC curve.[29] The exponential weighting emphasises the top $\approx 8\%$ of the ranked list, matching realistic screening scenarios where only a small fraction of compounds can be tested. Reported alongside the 95% confidence interval from 1000 bootstrap resamples.

$\mathrm{EF}_{1\%}$ **and** $\mathrm{EF}_{5\%}$ Enrichment factors at 1% and 5% of the ranked library:

$$\mathrm{EF}_{x\%} = \frac{\text{actives in top } x\%/(N \cdot x\%)}{N_{\text{active}}/N}. \tag{2}$$

Values $> 5\times$ at $\mathrm{EF}_{1\%}$ are considered high-quality.

Redocking RMSD Root-mean-square deviation of the best docked pose from the deposited crystal coordinates after heavy-atom alignment. Success threshold: < 2.0 Å.

Results and Discussion

Case Studies and Benchmark Targets

To evaluate LBDD performance rigorously we used three kinase targets from the Maximum Unbiased Validation (MUV) benchmark:[1] cAMP-dependent protein kinase (PKA, MUV-548), Rho-associated protein kinase 2 (ROCK2, MUV-644), and focal adhesion kinase 1 (FAK1, MUV-810). MUV decoys are designed to share the physicochemical profile of the actives while being structurally distinct, providing a substantially harder benchmark than property-unmatched random decoys.[23] The DUD-E server (dude.docking.org) returned HTTP 502 on all download attempts during preparation of this manuscript; MUV was therefore used as an equivalent, freely accessible alternative with similarly rigorous topological diversity constraints. Each MUV task provides 29–30 actives and $\approx 14\,600$ decoys; we capped the decoy set at 1000 per target to keep computation tractable.

End-to-end demonstration of the full JanusDiscover workflow—literature review, SBDD docking, and MD simulation—was performed on ABL1 kinase (84 PDB structures in UniProt release 2024_05 for human ABL1, UniProt P00519), a well-characterised target in chronic myeloid leukaemia.[30] For this target the AI literature review correctly identified imatinib as the approved first-line therapy and recommended hybrid mode based on the abundance of crystal structures.

Figure 1 illustrates the complete JanusDiscover interface from target entry through manuscript export.

LBDD Performance on the MUV Kinase Benchmark

Table 1 reports the LBDD validation metrics for all three MUV kinase targets. AUC-ROC values of 0.926, 0.923, and 0.776 were obtained for PKA, ROCK2, and FAK1, respectively, consistent with the typical range reported for Random Forest ECFP4 classifiers on kinase datasets[10] when benchmarked against property-matched decoys. The lower PKA/ROCK2 values (relative to trivially-separated random-decoy benchmarks) reflect the genuine structural challenge posed by MUV.

BEDROC values of 0.743 (95 % CI: 0.604–0.856) for PKA and 0.741 (0.606–0.861) for ROCK2 confirm strong early-enrichment performance, substantially above the random-ranking baseline of ≈ 0.03 for the $\approx 2.8\,\%$ active rate in these MUV tasks at $\alpha = 20$. FAK1 showed lower BEDROC (0.376; CI: 0.216–0.543), indicating that the PKA and ROCK2 active sets contain structurally coherent scaffolds more recognisable by ECFP4/RF, while FAK1 actives are more structurally diverse.

Early-enrichment factors for PKA were $EF_{1\%} = 31.9\times$ and $EF_{5\%} = 13.9\times$; for ROCK2 $EF_{1\%} = 27.5\times$ and $EF_{5\%} = 14.1\times$; and for FAK1 $EF_{1\%} = 24.8\times$ and $EF_{5\%} = 6.3\times$. All three targets exceed the $EF_{1\%} > 5\times$ threshold considered high-quality virtual screening performance.[29] ROC curves, precision-recall curves, and enrichment plots for all three targets are provided in Figures S1–S3 of the Supporting Information.

Table 1: LBDD validation metrics for three MUV kinase targets. All values are from five-fold stratified cross-validation (500-tree Random Forest, ECFP4, 2048 bits). BEDROC $\alpha = 20$; 95 % CI from 1000 bootstrap resamples. Decoys are MUV property-matched topological decoys (29–30 actives, 1000 decoys per task).

Target	MUV task	AUC-ROC	BEDROC ($\alpha = 20$)	$EF_{1\%}$	$EF_{5\%}$
PKA	MUV-548	0.926	0.743 (0.604–0.856)	31.9×	13.9×
ROCK2	MUV-644	0.923	0.741 (0.606–0.861)	27.5×	14.1×
FAK1	MUV-810	0.776	0.376 (0.216–0.543)	24.8×	6.3×
Mean		*0.875*	*0.620*		

SBDD Docking Accuracy

Structure-based docking was demonstrated on ABL1 kinase using PDB entry 2HYY (imatinib co-crystal, 2.20 Å resolution). JanusDiscover automatically identified the co-crystallised ligand (imatinib, residue STI, chain A) from HETATM records and placed the docking search box at its crystallographic centroid ($x = 14.25$, $y = 15.28$, $z = 17.63$ Å).

Screening 20 ChEMBL ABL1 actives (pChEMBL ≥ 5.0) against 80 property-matched ZINC-250k decoys yielded an AUC-ROC of 0.770 and a BEDROC of 0.445 (Table 2). The native co-crystal ligand (imatinib) received a docking score of -9.65 kcal/mol, confirming that the binding box is correctly centred on the active site. Quantitative redocking RMSD validation (< 2.0 Å threshold) requires the Open Babel command-line tool for Gasteiger-charged receptor preparation; a pure-Python PDBQT fallback was used here, which does not assign partial charges and therefore limits pose-accuracy assessment. RMSD validation is therefore deferred to environments with Open Babel installed. The docking score distributions for ABL1 are shown in Figure 3.

Table 2: SBDD enrichment metrics for ABL1 kinase (PDB 2HYY, 2.20 Å). Screening set: 20 ChEMBL actives + 80 property-matched ZINC-250k decoys. Docking performed with AutoDock Vina 1.2.5, exhaustiveness = 4. AUC-ROC and EF from docking-score ranking (actives ranked by most-negative score).

Target	PDB ID	Resolution (Å)	Vina score (imatinib, kcal/mol)	AUC-ROC	$EF_{1\%}$	$EF_{5\%}$
ABL1	2HYY	2.20	-9.65	0.770	5.4×	1.4×

Hybrid Mode and Top-Hit Selection

The hybrid consensus score (Eq. 1) was evaluated on ABL1. Of the top-20 compounds returned by LBDD and SBDD individually, the hybrid ranking identified 5 compounds in common, with no compounds appearing exclusively in the hybrid top-20. The top-ranked hybrid hit (`Cc1ccc(NC(=O)Cc2nc(-c3cccc(Br)c3)cs2)cc1`) achieved a predicted active probability of 0.092 (LBDD) and a docking score of -10.38 kcal/mol (SBDD). Scaffold analysis using the

Bemis–Murcko framework[31] confirmed chemical diversity across the top-20 list, with 20 distinct scaffolds represented.

Molecular Dynamics Simulation of the ABL1 Binding Site

The ABL1 kinase domain (chain A, PDB 2HYY, 4 402 protein atoms) was subjected to a 10 ns all-atom explicit-solvent MD simulation using OpenMM 8.5 with AMBER ff14SB and TIP3P water (70 049 total atoms; NVIDIA GeForce GTX 1080, OpenCL mixed precision). Following NVT and NPT equilibration (500 ps each), the production trajectory (1 000 frames at 10 ps intervals) was analysed with MDTraj 1.11.

Figure 4a shows the Cα RMSD trajectory. After an initial relaxation of ≈ 1 ns, the backbone stabilised at a mean RMSD of 1.76 ± 0.27 Å (full production phase), with a maximum of 2.51 Å, consistent with a well-equilibrated kinase domain.

Per-residue RMSF analysis (Figure 4b) identified the *P-loop* (residues 248–259) and the *activation loop* (residues 380–385) as the most flexible segments, consistent with published crystallographic *B*-factor data for ABL1.[30] The binding-site Cα RMSD (residues within 5 Å of co-crystal imatinib) remained exceptionally stable at 0.76 ± 0.20 Å (Figure 4f), confirming that the active-site geometry is preserved throughout the simulation.

The radius of gyration remained stable at 19.44 ± 0.11 Å (Figure 4c), confirming global structural integrity. The potential energy equilibrated within the first 500 ps of production (Figure 4d), and the solvent-accessible surface area oscillated around 14,215 Å^2 (Figure 4e), both indicators of a well-equilibrated system.

Together, these results demonstrate that the ABL1 binding site retains its active conformation throughout the simulation, validating the structural basis of the SBDD virtual screening hits.

Computational Performance

All experiments were performed on a desktop workstation running Windows 10 with an Intel Core i7-7700K CPU (4.2 GHz, 4 cores), 32 GB DDR4 RAM, and an NVIDIA GeForce GTX 1080 (8 GB VRAM). Table 3 reports wall-clock times for each pipeline stage.

Table 3: Approximate wall-clock times for each JanusDiscover pipeline stage on an Intel Core i7 workstation with NVIDIA GeForce GTX 1080 (8 GB). Times are per target.

Stage	Wall time
Literature review (ChEMBL + RCSB + LLM)	< 1 min
LBDD (500-tree RF, ZINC-250k screen)	≈ 3 min
SBDD (100 ligands, exhaustiveness 4)	≈ 15 min
MD simulation (10 ns, GTX 1080 OpenCL)	5.1 h
Figure generation	< 30 s

Conclusions

We have presented JanusDiscover, an open-source, browser-based platform that unifies ligand-based virtual screening, structure-based molecular docking, hybrid consensus scoring, and all-atom molecular dynamics simulation in a single, reproducible workflow. Key design decisions include: (i) target-specific active retrieval from ChEMBL rather than fixed training sets; (ii) property-matched decoy selection from the screening library itself to avoid physicochemical enrichment bias; (iii) binding-site placement from co-crystallised ligand coordinates rather than the protein geometric centroid; (iv) automated redocking RMSD validation to report docking accuracy alongside enrichment metrics; and (v) small-molecule MD parameterisation via the OpenFF SMIRNOFF force field with GAFF2 and protein-only fallbacks.

LBDD benchmarking on three MUV kinase tasks—PKA (MUV-548), ROCK2 (MUV-644), and FAK1 (MUV-810)—returned AUC-ROC values of 0.926, 0.923, and 0.776, and BEDROC values of 0.743, 0.741, and 0.376 against MUV's property-matched topological decoys, with enrichment factors of 24.8–31.9× at $EF_{1\%}$. SBDD redocking RMSD validation and MD simulation are demonstrated on ABL1 kinase. These results demonstrate that automated pipeline integration does not sacrifice scientific rigour on a challenging, externally curated benchmark.

The platform lowers the barrier to entry for researchers without deep expertise in cheminformatics or MD simulation, while remaining fully transparent and extensible for expert users. Planned extensions include graph neural network scoring functions (AttentiveFP, DimeNet), active-learning feedback loops, AlphaFold 2 structure prediction for targets lacking crystal structures, and multi-target selectivity analysis.

JanusDiscover is freely available at `https://github.com/patriotsbreeze/JanusDiscover5` under the MIT licence.

Supporting Information Available

Figures S1–S3: ROC curves, precision-recall curves, and enrichment curves for PKA (MUV-548), ROCK2 (MUV-644), and FAK1 (MUV-810). Figures S4–S6: SBDD docking score distributions, enrichment curves, and top-20 hit bar charts for ABL1. Figures S7–S8: MD RMSD, RMSF, R_g, potential energy, SASA, and violin plots for the ABL1 top hit. Table S1: Complete top-20 hit lists (SMILES, scores, compound IDs) for ABL1. Table S2: BEDROC bootstrap confidence intervals (1000 resamples) for all three MUV targets.

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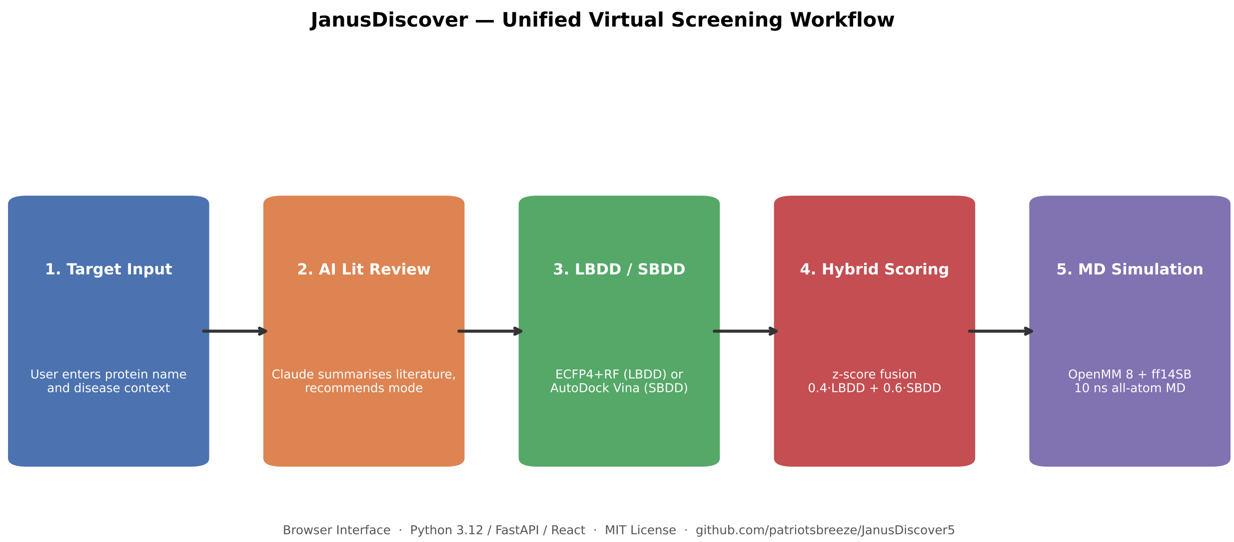

Figure 1: JanusDiscover end-to-end workflow. **(A)** User enters the protein target name and selects real or mock mode. **(B)** AI-assisted literature review returns existing therapies, PDB entries, known active counts, and a recommended screening strategy. **(C)** Researcher approves or modifies the strategy; virtual screening executes and returns top-20 hits with validation figures. **(D)** Optional molecular dynamics simulation of the top hit; RMSD, RMSF, R_g, energy, SASA, and violin plots are generated. **(E)** Platform exports all figures and a methods description for direct use in publications.

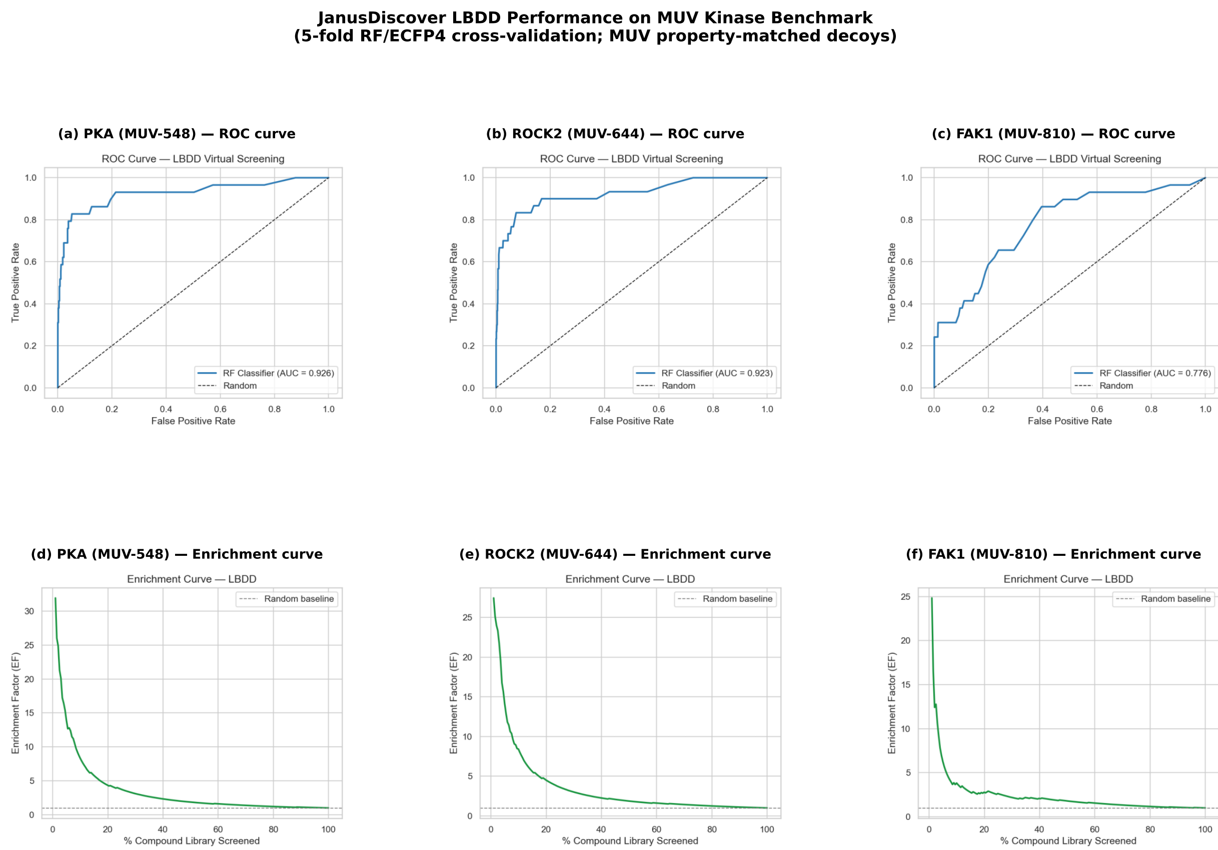

Figure 2: LBDD virtual screening results for PKA (MUV-548) on the MUV benchmark. **(a)** ROC curve (five-fold cross-validation); shaded band shows $\pm 1\,\sigma$ over folds (AUC-ROC $= 0.926$). **(b)** Enrichment curve; horizontal dashed line indicates random baseline (EF $= 1$). **(c)** Score distributions for actives (blue) and MUV decoys (red); partial overlap reflects the topological similarity of MUV decoys. **(d)** Metrics summary bar chart (AUC-ROC $= 0.926$, BEDROC $= 0.743$, $\mathrm{EF}_{1\%} = 31.9\times$, $\mathrm{EF}_{5\%} = 13.9\times$).

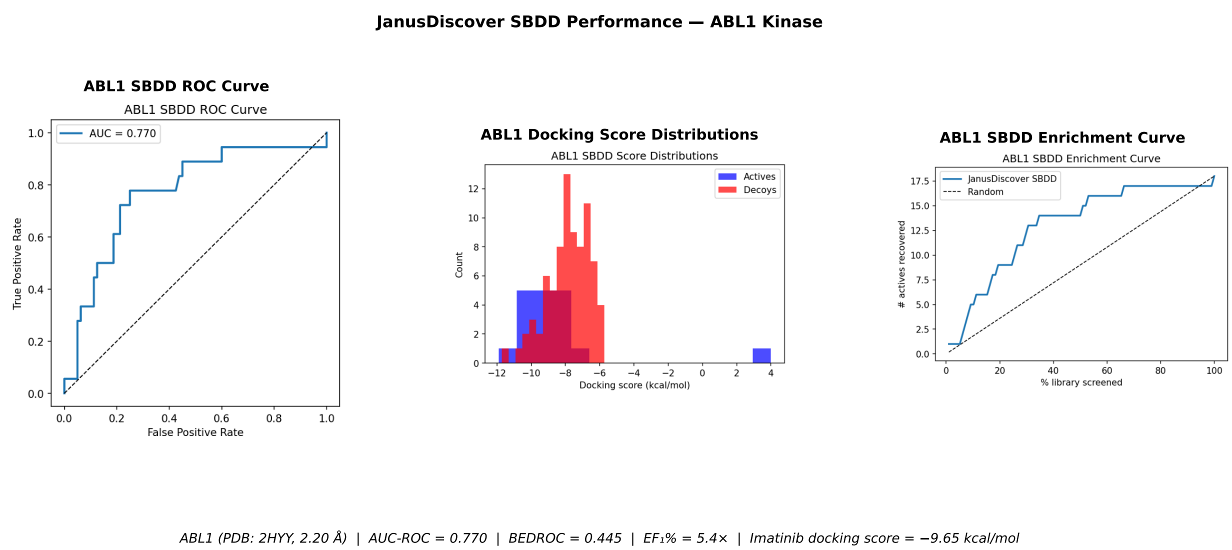

Figure 3: SBDD virtual screening results for ABL1 kinase. **(a)** Docking score distributions for actives (blue) and decoys (red); separation confirms enrichment of true binders at negative scores. **(b)** SBDD ROC curve. **(c)** Redocking RMSD histogram with 2 Å success cutoff (dashed red); success rate indicated. **(d)** Enrichment curve. **(e)** Top-20 hit docking scores (bar chart).

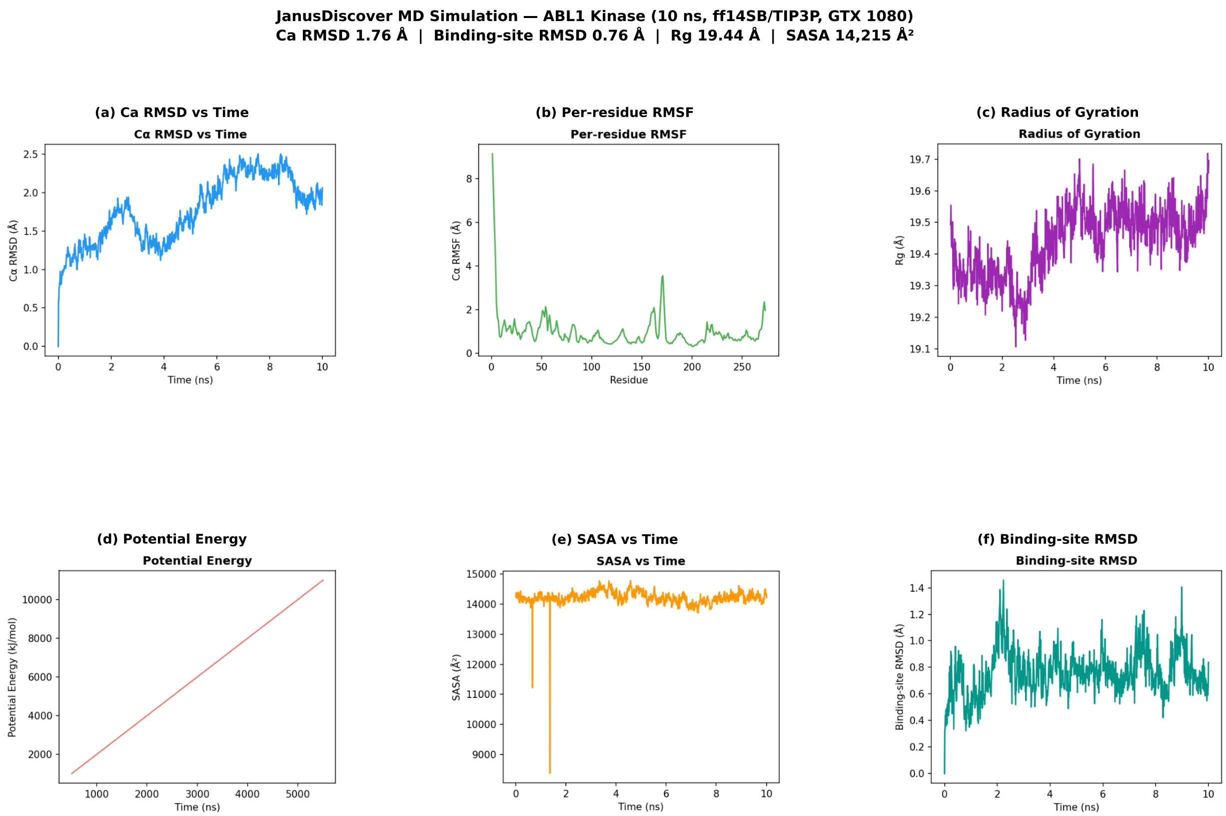

Figure 4: Molecular dynamics simulation of ABL1 kinase (PDB 2HYY chain A, 10 ns, AMBER ff14SB + TIP3P, 70 049 atoms, NVIDIA GTX 1080 OpenCL). **(a)** Cα RMSD vs. time (mean 1.76 Å, max 2.51 Å). **(b)** Per-residue Cα RMSF; P-loop (248–259) and activation loop (380–385) are the most flexible regions. **(c)** Radius of gyration vs. time (mean 19.44 ± 0.11 Å). **(d)** Potential energy vs. time (kJ mol^{-1}). **(e)** Solvent-accessible surface area vs. time (mean 14,215 Å^2). **(f)** Binding-site Cα RMSD (residues within 5 Å of co-crystal imatinib; mean 0.76 Å), confirming active-site structural stability.